AWS for SAP

Voice of Customer Analytics using SAP Data Intelligence on AWS

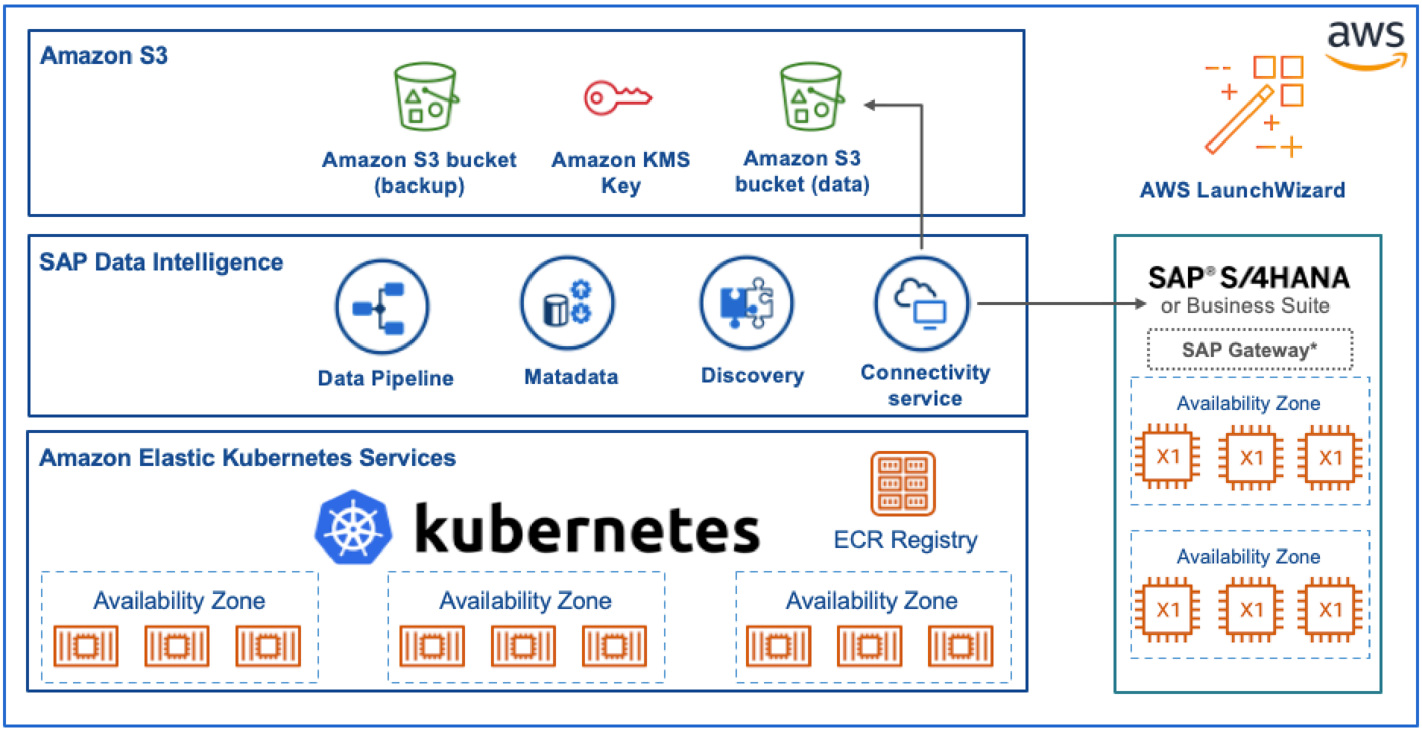
by Patrick Leung and Pavol Masarovic | on 15 FEB 2021 | in Amazon Elastic Kubernetes Service, AWS Cloud Development Kit, SAP on AWS | Permalink | Share

Introduction

Most companies have processes to better understand their customers and data is fast becoming the key to strengthening customer relationships. However, merely collecting data isn't enough. While most organizations have access to plenty of data, determining the meaning behind the data can be difficult. In this blog post, we show how you can use a combination of SAP Data Intelligence 3 (DI3), SAP HANA and Amazon S3 to build a data-driven architecture to better understand customer feedback and derive meaningful insight.

Overview

We start with the deployment of SAP Data Intelligence 3 and show how easy it is to export data from an Amazon S3 data lake into SAP HANA using SAP Data Intelligence 3. Then we show how to use SAP HANA to derive meaningful insight from textual data. Once we have finished with the analysis, these data are then imported back in to Amazon S3 for long-term low-cost storage.



SAP Data Intelligence and SAP HANA deployment on AWS

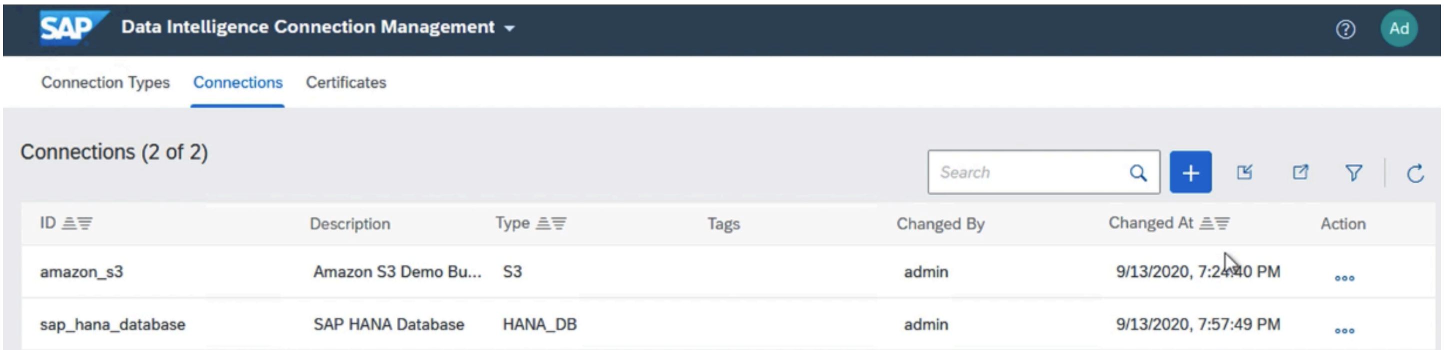
Let's start with a quick introduction of SAP Data Intelligence (DI3). Most readers probably recognize the product's predecessor, SAP Data Hub. From a technical perspective, this is completely different from the traditional SAP NetWeaver architecture and runs entirely on Kubernetes. As this runs in Kubernetes, you can deploy this very quickly using the AWS Cloud Development Kit (CDK) and Amazon Elastic Kubernetes Service (Amazon EKS). There are three key steps in the deployment process.

1. AWS CDK deployment of Amazon EKS
2. DI3 installation using Software Life-cycle Container (SLC) bridge on Kubernetes in unattended mode.
3. Deployment of SAP HANA. The easiest way to deploy this is using AWS LaunchWizard for SAP.

Follow the installation guide in AWS Sample GitHub for step 1 and step 2 and AWS LaunchWizard for SAP deployment guide for step 3.

SAP Data Intelligence Connection Management

Let's take a quick look at the connection management and see how easy it is to set up a connection to Amazon S3 . Amazon S3 is fully integrated into SAP DI3. Simply enter your AWS access key and secret access key together with the bucket name and AWS region. That's it!



Setting up the HANA database connection is very easy if both the SAP Data Intelligence and SAP HANA run in the same Virtual Private Cloud (VPC) on AWS. All you need is the SAP HANA database host name, port and the username and password with the required security group opened.

SAP Data Intelligence Pipeline Modeller

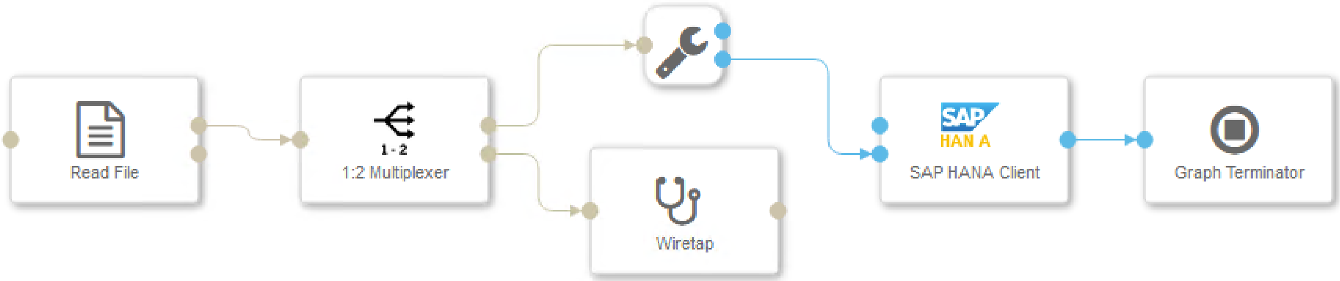
Now that we have set up the connection, we show you how to create an example pipeline to extract data from Amazon S3 in to SAP HANA. The SAP DI3 modeler application is based on the pipeline engine that uses a flow-based programming paradigm to create data processing pipelines. The modeler also provides predefined operators that integrates into Amazon SNS, Amazon Redshift and Amazon S3.

Import the sample graph from the AWS sample GitHub. Once the graph has been imported, change the **SAP HANA Client** and **READ FILE** operator in the graph. The source data we are going to use is from Amazon Customer Reviews Dataset. This is publicly available and you can use the command below to download a sample set of data directly.

Bash

```
aws s3 cp s3://amazon-reviews-ml/json/dev/dataset_en_dev.json <Amazon S3 Location>
```

The imported graph will look like this.



You can start the job by clicking on the **Run** button. This takes a few minutes to complete and the example dataset is shown below.

```
SELECT TOP 1000 * FROM "SAPA4H"."$TA_SENTIMENTINDX"
```

	REVIEWID	TA_RULE	TA_COUNTER	TA_TOKEN	TA_LANGUAGE	TA_TYPE	TA_NORMALIZED	TA_STEM	TA_PARAGRAPH	TA_SENTENCE	TA_CREATED_AT	TA_OFFSET	TA_PARENT
1	RDIJS7QYB6XNR	Entity Extraction	1	Excellent	en	Sentiment	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	0	?
2	RDIJS7QYB6XNR	Entity Extraction	2	Excellent	en	StrongPositiv...	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	0	1
3	R298788GS6I901	Entity Extraction	1	my daughter	en	Topic	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	0	2
4	R298788GS6I901	Entity Extraction	2	my daught...	en	Sentiment	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	0	?
5	R298788GS6I901	Entity Extraction	3	loved	en	StrongPositiv...	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	12	2
6	R298788GS6I901	Entity Extraction	4	liked	en	WeakPositive...	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	27	5
7	R298788GS6I901	Entity Extraction	5	liked the pri...	en	Sentiment	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	27	?
8	R298788GS6I901	Entity Extraction	6	the price	en	Topic	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	33	5
9	R298788GS6I901	Entity Extraction	7	Amazon	en	ORGANIZATI...	?	?	1	2	Dec 6, 2020 8:37:53.0 PM	114	?
10	R298788GS6I901	Entity Extraction	8	Best	en	Sentiment	?	?	1	2	Dec 6, 2020 8:37:53.0 PM	128	?
11	R298788GS6I901	Entity Extraction	9	Best	en	StrongPositiv...	?	?	1	2	Dec 6, 2020 8:37:53.0 PM	128	8
12	R3BPETL222L...	Entity Extraction	1	Great	en	StrongPositiv...	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	0	2
13	R3BPETL222L...	Entity Extraction	2	Great item	en	Sentiment	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	0	?
14	R3BPETL222L...	Entity Extraction	3	item	en	Topic	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	6	2
15	R3BPETL222L...	Entity Extraction	4	Pictures pop	en	NOUN_GROUP	?	?	1	2	Dec 6, 2020 8:37:53.0 PM	12	?
16	R3SORMPJZO3...	Entity Extraction	1	crazy glue	en	NOUN_GROUP	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	29	?
17	R2RDOJQ0WB...	Entity Extraction	1	pleased	en	WeakPositive...	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	6	2
18	R2RDOJQ0WB...	Entity Extraction	2	pleased wit...	en	Sentiment	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	6	?
19	R2RDOJQ0WB...	Entity Extraction	3	the product	en	Topic	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	19	2
20	RNX4EXOBBPN5	Entity Extraction	1	Do not buy	en	Sentiment	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	0	?
21	RNX4EXOBBPN5	Entity Extraction	2	Do not buy	en	StrongNegati...	?	?	1	1	Dec 6, 2020 8:37:53.0 PM	0	1
22	RNX4EXOBBPN5	Entity Extraction	3	break	en	Sentiment	?	?	1	2	Dec 6, 2020 8:37:53.0 PM	23	?
23	RNX4EXOBBPN5	Entity Extraction	4	break	en	MajorProblem	?	?	1	2	Dec 6, 2020 8:37:53.0 PM	23	3
24	RNX4EXOBBPN5	Entity Extraction	5	15 minutes	en	TIME_PERIOD	?	?	1	2	Dec 6, 2020 8:37:53.0 PM	55	?
25	RNX4EXOBBPN5	Entity Extraction	6	waste your ...	en	Sentiment	?	?	1	2	Dec 6, 2020 8:37:53.0 PM	93	?
26	RNX4EXOBBPN5	Entity Extraction	7	waste	en	NeutralSenti...	?	?	1	2	Dec 6, 2020 8:37:53.0 PM	93	6
27	RNX4EXOBBPN5	Entity Extraction	8	your money	en	Topic	?	?	1	2	Dec 6, 2020 8:37:53.0 PM	99	6
28	RNX4EXOBBPN5	Entity Extraction	9	cheap plastic	en	NOUN_GROUP	?	?	1	3	Dec 6, 2020 8:37:53.0 PM	130	?
29	RNX4EXOBBPN5	Entity Extraction	10	poi balls	en	NOUN_GROUP	?	?	1	4	Dec 6, 2020 8:37:53.0 PM	177	?

Now the data is in SAP HANA database, let's use the text analysis feature in SAP HANA to understand the data set better. First, we will create an primary key for the table named `REVIEWID`

SQL

```sql
ALTER TABLE <schema>.customer_reviews ADD PRIMARY KEY (REVIEWID)
```

Then we create an full text index called `SENTIMENTINDX` using the SAP HANA text analysis configuration mode – `EXTRACTION_CORE_VOICEOFCUSTOMER` for the reviewbody column. We are using this configuration because this includes a set of entity types and rules that address requirements for extracting customer sentiments and requests. There are a number of configuration modes such as `EXTRACTION_CORE_ENTERPRISE` and `EXTRACTION_CORE`. You can find the full list of configuration mode in the SAP HANA developer guide.

SQL

```
CREATE fulltext INDEX sentimentindx ON <schema>.customer_reviews ("REVIEWBODY") CONF
```

We can view the full text index using the command below.

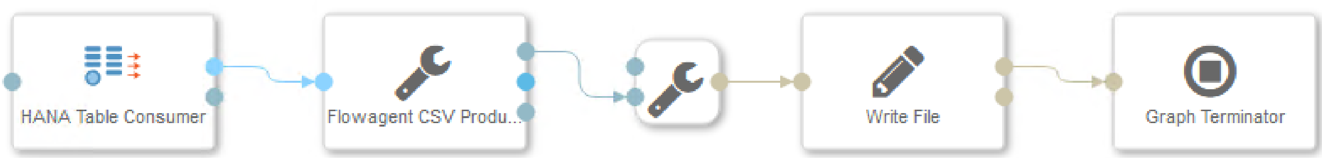
SQL

```
SELECT TOP 1000 * FROM <schema>."$TA_SENTIMENTINDX"
```

	REVIEWID	TA_RULE	TA_COUNTER	TA_TOKEN	TA_LANGUAGE	TA_TYPE
1	RDIJS7QYB6XNR	Entity Extraction	1	Excellent	en	Sentiment
2	RDIJS7QYB6XNR	Entity Extraction	2	Excellent	en	StrongPositiveSentiment
3	R298788GS6I901	Entity Extraction	1	my daughter	en	Topic
4	R298788GS6I901	Entity Extraction	2	my daughter loved	en	Sentiment
5	R298788GS6I901	Entity Extraction	3	loved	en	StrongPositiveSentiment
6	R298788GS6I901	Entity Extraction	4	liked	en	WeakPositiveSentiment
7	R298788GS6I901	Entity Extraction	5	liked the price	en	Sentiment
8	R298788GS6I901	Entity Extraction	6	the price	en	Topic
9	R298788GS6I901	Entity Extraction	7	Amazon	en	ORGANIZATION/COMMERCIAL
10	R298788GS6I901	Entity Extraction	8	Best	en	Sentiment
11	R298788GS6I901	Entity Extraction	9	Best	en	StrongPositiveSentiment
12	R3BPETL222L...	Entity Extraction	1	Great	en	StrongPositiveSentiment
13	R3BPETL222L...	Entity Extraction	2	Great item	en	Sentiment
14	R3BPETL222L...	Entity Extraction	3	item	en	Topic
15	R3BPETL222L...	Entity Extraction	4	Pictures pop	en	NOUN_GROUP
16	R3SORMPJZO...	Entity Extraction	1	crazy glue	en	NOUN_GROUP
17	R2RDOJQ0WB...	Entity Extraction	1	pleased	en	WeakPositiveSentiment
18	R2RDOJQ0WB...	Entity Extraction	2	pleased with the product	en	Sentiment
19	R2RDOJQ0WB...	Entity Extraction	3	the product	en	Topic
20	RNX4EXOBBP...	Entity Extraction	1	Do not buy	en	Sentiment
21	RNX4EXOBBP...	Entity Extraction	2	Do not buy	en	StrongNegativeSentiment
22	RNX4EXOBBP...	Entity Extraction	3	break	en	Sentiment
23	RNX4EXOBBP...	Entity Extraction	4	break	en	MajorProblem
24	RNX4EXOBBP...	Entity Extraction	5	15 minutes	en	TIME_PERIOD

The text analysis recognized that the `TA_TOKEN` "Excellent", "Great" and "Best" are strong positive sentiment, "Do not buy" is a strong negative sentiment and "my daughter", "the price" and 'item" are topic. The `TA_LANGUAGE` column indicates the language of the document and this can be very useful as customer reviews can come in many different languages.

Once the analysis has been completed, you can move the full-text index into Amazon S3 for lower cost long term storage. This pipeline uses a HANA table consumer operator and converts it into a CSV, then use the write file operator to write to Amazon S3.



Conclusion

In this blog post, we show how easy it is to move data between Amazon S3 and SAP HANA using SAP Data Intelligence 3. This is just a glimpse of what is possible with SAP on AWS. SAP's Intelligent Enterprise portfolio and AWS Cloud services are enabling enterprises to create new business models faster. If you have questions or would like to know about SAP on AWS innovations, please contact the SAP on AWS team using this link or visit aws.com/sap to learn more. Start building on AWS today and have fun!